SRM Institute of Science and Technology
College of Engineering and Technology
School of Computing
(Common to all Branches)

SRM Nagar, Kattankulathur – 603203, Chengalpattu District, Tamilnadu

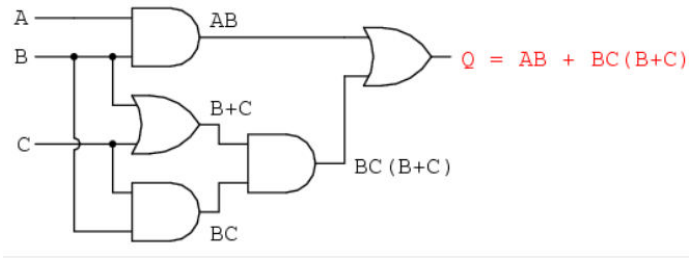
Academic Year: 23-24/ODD

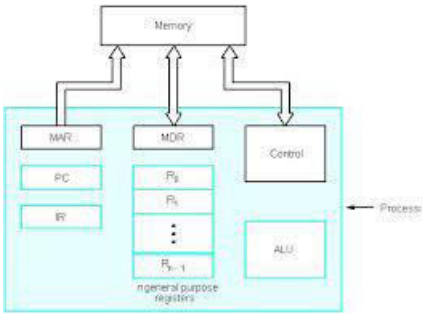
Mode of Exam
OFFLINE

Test: CLAT-1 **Date: 18/08/2023**
Course Code & Title: 21CSS201T – Computer Organization and Architecture **Duration:** 2 periods
Year & Sem: II Yr, III sem SET B **Max. Marks:** 50 marks

Q. No	Question
1	The given hexadecimal number (1E.53)16 is equivalent to _____ a) (35.684)8 **b) (36.246)8** c) (34.340)8 d) (35.599)8
2	The number of characters that can be represented in ASCII-8 are _____ a) 128 **b) 256** c) 32 d) 64
3	**What would be the gray code equal to the number 14?** **a. 1000** **b. 1110** **c. 1111** d. 1001
4	**The 2's complement of binary number 1010 is _____** **a.** 0101 **b.** 1010 **c.** **0110** **d.** 1110
5	**Truth Table:** Input A | Input B | Output $Y = \overline{A \oplus B}$ 0 | 0 | 1 0 | 1 | 0 1 | 0 | 0 1 | 1 | 1 The above truth table refers to a. **XNOR**

	b. NOR c. XOR d. NAND
6	**___ is the decimal based equivalent form of binary number 0.0111?** **Ans: 0.43750(Option is not given below so give 1 mark)** **a.** -0.8750 **b.** 0. 3750 c. 0. 8750 **d.** 8.750
7	**. The sign magnitude representation of -9 is _____** a)00001001 b)11111001 **c)10001001** d) 11001
8	**Which of the following is the BCD representation of 357?** **a. 0011 0101 0111** **b.** 1011 0101 0111 **c.** 0011 0111 0111 a. 0011 0101 1111
9	**Consider a CPU that has 64- bit, the address of the contents are 1000, 1008,1016, 1024. Now MAR is holding an address is 1008 to fetch the content from the memory. Analyze the operation and select the following address of the content which is held by the Program Counter (PC).** a. 1000 b. 1008 **c. 1016** d. 1024
10	**If the data "0010" is to be stored in a little-endian machine, it will be stored as** a. 0010 b. 0100 c. 1000 d. 0001
11	**Add the following signed binary number** a. 111000 + 001101 000101 b. 11001100 + 00010010 11011110

	c. 111100001111 + 001100001101 001000011100 d. 11000011 + 00011000 11011011
12	**Draw the circuit diagram for the following logic expression AB + BC(B + C)** 
13	**Convert the given binary string to gray code** i) 011001=010101 i) 110011=101010
14	**Convert decimal (62)10 to Excess-3** 10010101
15	**List out the drawbacks of single bus structure** The devices connected to a bus vary widely in their speed of operation. Some devices are relatively slow, such as printer and keyboard. Some devices are considerably fast, such as optical disks. Memory and processor units operate are the fastest parts of a computer.
20	**Perform the following conversion** (i) $(176.35)_8$ to Binary 001111110.011101 (ii) $(1011100011001.01)_2$ to decimal 5913.5 iii)$(7892)_{10}$ to hexadecimal 1ED4 iv) **Perform BCD addition on the following numbers 652.21 and 345.12** **Addition: 100110010111.00110011** (OR) b. i) **Subtract (23) from (-19) using 2's complement notation** Ans: 0101010 ii) **Find the multiplication of 01101 and 01011?** 010001111

	iii) Divide 20/4 Q=11 R=00
21	**i) Convert decimal (61.81)10 to excess-3.** 10010100.11000100 **ii) Convert (00110110)2 to Excess-3** 01101001 **iii) Convert Excess-3 (36) to binary** **00000011** **(OR)** i) Using the neat diagram, what are the main elements of the operational idea of computer organization write in detail?  The operational concept of computer organization refers to how a computer system is designed and structured to perform its tasks efficiently and effectively. It encompasses various aspects such as the organization of hardware components, memory hierarchy, instruction execution cycle, and the interaction between different components to execute programs. Here are the key components of the operational concept of computer organization: a. Instruction Execution Cycle: This is the fundamental process by which a computer executes instructions. It typically involves fetching an instruction from memory, decoding it to understand the operation to be performed, fetching the operands if necessary, executing the operation, and storing the result back in memory or a register. b. Memory Hierarchy: Computers use a memory hierarchy that includes different types of memory with varying speeds and capacities. This hierarchy includes registers (fastest but smallest), cache memory (faster but limited in size), main memory (slower but larger), and secondary storage (even slower but with much larger capacity). The memory hierarchy aims to provide a balance between speed and cost. c. Central Processing Unit (CPU): The CPU is the heart of the computer and is responsible for executing instructions. It contains the Arithmetic Logic Unit (ALU) for performing arithmetic and logical operations, and the Control Unit (CU) for managing the execution of instructions and coordinating data movement within the system. d. Instruction Set Architecture (ISA): The ISA defines the set of instructions that a computer can execute. It includes the instruction format, addressing modes, data types, and operations supported by the CPU. The ISA serves as an interface between software and hardware. e. Pipelining and Parallelism: To improve performance, modern CPUs often use pipelining and various forms of parallelism. Pipelining divides the instruction

execution cycle into stages, allowing multiple instructions to be processed concurrently at different stages. Parallelism involves executing multiple instructions simultaneously to increase overall throughput.

f. Input/Output (I/O) Operations: Computers interact with the outside world through input and output operations. These operations involve transferring data between the computer's internal memory and external devices such as keyboards, displays, storage devices, and network interfaces.

g. Interrupts and Exception Handling: Interrupts are signals that interrupt the normal execution of a program to handle external events, such as hardware device requests or errors. Exception handling involves managing unexpected events, errors, and exceptional conditions during program execution.

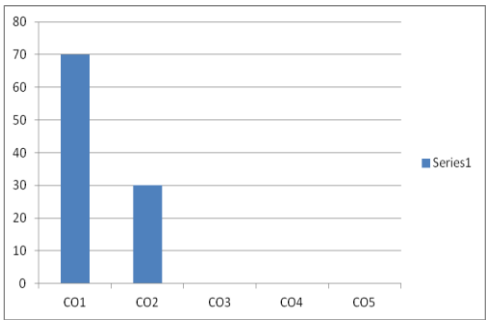
h. Bus Architecture: Computers use a bus system to facilitate communication between various components, including the CPU, memory, and I/O devices. The bus connects these components and allows them to exchange data and control signals.

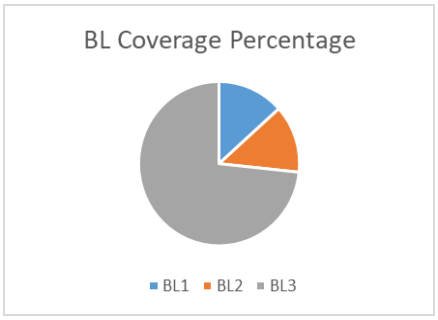
i. Clock and Synchronization: The computer's clock generates regular electrical pulses that synchronize the operations of various components. The clock speed influences the computer's overall performance, as it determines how many instructions can be executed in a given time frame.

j. Micro-architecture: Also known as the internal architecture or organization, micro-architecture refers to the specific design implementation of a CPU or other hardware component. It includes details about how instructions are executed, data paths, control signals, and pipeline stages.

***Performance Indicators are available separately for Computer Science and Engineering in AICTE examination reforms policy.**

Course Outcome (CO) and Bloom's level (BL) Coverage in Questions





Approved by the Audit Professor/Course Coordinator